

Econometrics I

The Multivariate Normal distribution

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Introduction and Goals

- For the multiple linear regression model under assumptions MLR1–MLR5 we now have considerable information about the sampling distributions of the least squares statistics $\hat{\beta}$ and $\hat{\mathbf{u}}'\hat{\mathbf{u}}$.
- That information, which concerns expectations, variances and covariances, is sufficient to justify the use of certain sample statistics and estimates of the population parameters and to provide an estimate of their precision.
- We have seen that $\hat{\beta}_j$ serves as an estimate of β_j and that $\hat{\sigma}_{\hat{\beta}_j} = \hat{\sigma} \sqrt{q^j j}$ serves as an estimate of its standard deviation $\sigma \sqrt{q^j j}$.
- However, we need more information to undertake further statistical inference, i.e., to construct confidence intervals and to test hypothesis at exact significance levels.
- In order to do this, we will need to make some extra assumptions about the error term in the linear regression model $\mathbf{u}$.
- To be able to describe these assumptions and to work with them, we will need to develop some theory about the multivariate normal distribution.
- In the Review of Probability and Statistics, we have encountered this, but we will need to dig deeper now, since you now possess the knowledge and techniques required to do it.

Multivariate normality

- Let $\mathbf{y}$ be a $n \times 1$ random vector, whose joint pdf is given by

$$f(\mathbf{y}) = (2\pi)^{-n/2} |\Sigma|^{-1/2} e^{-w/2}$$

where

- ▶ $w = (\mathbf{y} - \boldsymbol{\mu})' \Sigma^{-1} (\mathbf{y} - \boldsymbol{\mu})$ is a scalar.
 - ▶ $\boldsymbol{\mu}$ is a $n \times 1$ parameter vector to be interpreted as a vector of means.
 - ▶ Σ is a $n \times n$ positive definite parameter matrix to be interpreted as the variance-covariance matrix.
 - ▶ $|\Sigma|^{-1/2} = 1/\sqrt{\det(\Sigma)}$ where $\det$ denotes the determinant.
- Then we say that the distribution of $\mathbf{y}$ is multivariate normal or multinormal and $\mathbf{y} \sim N(\boldsymbol{\mu}, \Sigma)$
- Remark: notice that the scalar w is actually a **quadratic form**.

Two special cases

- Consider the case $n = 1$, then $\mathbf{y} = y$, $\boldsymbol{\mu} = \mu$ and $\boldsymbol{\Sigma} = \sigma^2$, $w = z^2$, with $z = (y - \mu)/\sigma$. So $f(y)$ is the familiar normal density: $f(y) = \exp(-z^2/2)/\sqrt{2\pi\sigma^2}$.
- Consider the case $n = 2$ then, $\mathbf{y} = (y_1, y_2)'$, $\boldsymbol{\mu} = (\mu_1, \mu_2)$, $\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$.
- Thus $\det(\boldsymbol{\Sigma}) = \sigma_{11}\sigma_{22} - \sigma_{12}^2 = \sigma_1^2\sigma_2^2(1 - \rho^2)$ where $\rho = \sigma_{12}/(\sigma_1\sigma_2)$, $\sigma_1^2 = \sigma_{11}$ and $\sigma_2^2 = \sigma_{22}$ and $\boldsymbol{\Sigma}^{-1} = (1 - \rho^2)^{-1} \begin{bmatrix} 1/\sigma_1^2 & -\rho/(\sigma_1\sigma_2) \\ -\rho/(\sigma_1\sigma_2) & 1/\sigma_2^2 \end{bmatrix}$
- Also $w = (z_1^2 + z_2^2 - 2\rho z_1 z_2)/(1 - \rho^2)$, with $z_1 = (y - \mu_1)/\sigma_1$, $z_2 = (y - \mu_2)/\sigma_2$.
- Some remarks are in order: σ_1^2 and σ_2^2 are the variances of y_1 and y_2 , σ_{12} is the covariance between y_1 and y_2 , and ρ is the correlation coefficient. In this case $f(y_1, y_2)$ follows a standard parameterization of the bivariate normal, in terms of its means, standard deviations and correlation coefficient.

Properties of a multinormal vector

- Return to the general case, partition the $n \times 1$ vector $\mathbf{y}$ into a $n_1 \times 1$ vector $\mathbf{y}_1$ and a $n_2 \times 1$ vector $\mathbf{y}_2$.
- Do the same for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$:

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix}$$

- The following is a restatement, using matrix language, of properties we had reviewed before. If $\mathbf{y} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then:

P1: expectation and variance

The expectation and variance of $\mathbf{y}$ are respectively: $E(\mathbf{y}) = \boldsymbol{\mu}$ and $V(\mathbf{y}) = \boldsymbol{\Sigma}$.

P2: Normal Marginals

The marginal distribution of $\mathbf{y}_1$ is multinormal: $\mathbf{y}_1 \sim N(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_{11})$

P3: Conditional distribution is multinormal

The conditional distribution of $\mathbf{y}_2$ given $\mathbf{y}_1$ is multinormal: $\mathbf{y}_2|\mathbf{y}_1 \sim N(\boldsymbol{\mu}_2^*, \boldsymbol{\Sigma}_{22}^*)$ where

$$\boldsymbol{\mu}_2^* = E(\mathbf{y}_2|\mathbf{y}_1) = \boldsymbol{\alpha} + \mathbf{B}'\mathbf{y}_1,$$

$$\mathbf{B} = \boldsymbol{\Sigma}_{11}^{-1}\boldsymbol{\Sigma}_{12},$$

$$\boldsymbol{\alpha} = \boldsymbol{\mu}_2 - \mathbf{B}'\boldsymbol{\mu}_1,$$

$$\boldsymbol{\Sigma}_{22}^* = V(\mathbf{y}_2|\mathbf{y}_1) = \boldsymbol{\Sigma}_{22} - \mathbf{B}'\boldsymbol{\Sigma}_{11}\mathbf{B}$$

Remarks: Note that the conditional expectation function $E(\mathbf{y}_2|\mathbf{y}_1)$ is linear and that the conditional variance-covariance matrix is constant across $\mathbf{y}_1$.

Question: look at the expressions for $\mathbf{B}$ and $\boldsymbol{\alpha}$, do they remind you of something you have seen before?

P4: Uncorrelatedness implies independence

If $\Sigma_{12} = \mathbf{O}$, then $\mathbf{y}_1$ and $\mathbf{y}_2$ are independent random vectors.

Note that the roles of $\mathbf{y}_1$ and $\mathbf{y}_2$ can be reversed

Proof.

We can provide a proof of this fact that once I left just stated. If $\Sigma_{12} = \mathbf{0}$ then $\mathbf{B} = \mathbf{O}$ so $\boldsymbol{\mu}_2^* = \boldsymbol{\alpha} = \boldsymbol{\mu}_2$ and $\Sigma_{22}^* = \Sigma_{22}$. Thus $\mathbf{y}_2|\mathbf{y}_1 \sim N(\boldsymbol{\mu}_2, \Sigma_{22})$, for all $\mathbf{y}_1$. These conditional distributions are all the same and they coincide with the marginal distribution. Thus the vectors are stochastically independent. □

Special case 1: $n_1 = n_2 = 1$

This is the bivariate normal distribution, P2 and P3 Specialize respectively to $y_1 \sim N(\mu_1, \sigma_{11})$, $y_2|y_1 \sim N(\alpha + \beta y_1, \sigma^2)$, where $\beta = \sigma_{12}/\sigma_{11}$, $\alpha = \mu_2 - \beta\mu_1$, $\sigma^2 = \sigma_{11} - \beta^2\sigma_{11}$.

Special case 2: $n_1 = n - 1$ and $n_2 = 1$

Here y_2 is a scalar and $\mathbf{y}_1$ is a vector. The variables and parameters given this partition are

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_1 \\ y_2 \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \mu_2 \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\sigma}_{12} \\ \boldsymbol{\sigma}'_{12} & \sigma_{22} \end{bmatrix}$$

Here we have that $y_2|\mathbf{y}_1 \sim N(\alpha + \boldsymbol{\beta}'\mathbf{y}_1, \sigma^2)$ where $\boldsymbol{\beta} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\sigma}_{12}$, $\alpha = \mu_2 - \boldsymbol{\beta}'\boldsymbol{\mu}_1$, $\sigma^2 = \sigma_{22} - \boldsymbol{\beta}'\boldsymbol{\Sigma}_{11}\boldsymbol{\beta}$. Notice that this case satisfies the assumptions of a *multiple linear regression model*, i.e., the conditional expectation is linear $E(y_2|\mathbf{y}_1) = \alpha + \boldsymbol{\beta}'\mathbf{y}_1$ and the conditional variance $V(y_2|\mathbf{y}_1) = \sigma^2$ is constant.

P5: Linear functions of a multinormal vector are multinormal

If $\mathbf{z} = \mathbf{g} + \mathbf{H}\mathbf{y}$, where $\mathbf{g}$ and $\mathbf{H}$ are nonrandom, and $\mathbf{H}$ has full row rank, then $\mathbf{z} \sim N(\mathbf{g} + \mathbf{H}\boldsymbol{\mu}, \mathbf{H}\boldsymbol{\Sigma}\mathbf{H}')$.

- Why $\mathbf{H}$ needs to have full row rank? Remember that the row rank of a matrix is the number of *linearly independent rows*. This condition is required to rule out degeneracies, in order to ensure that $V(\mathbf{z})$ is nonsingular, which is a prerequisite for multinormality.
- Note also that if $\mathbf{y}$ is $n \times 1$ and $\mathbf{H}$ is $m \times n$ with $m > n$ then $\mathbf{H}$ cannot have full row rank. Thus, in general, we work with matrices $\mathbf{H}$ with $m < n$ and such that the row rank is equal to m . If we interpret n as the number of observations in a random sample, as in the applications we deal with, we can expect to have $m < n$.

Review: functions of a Standard Normal Vector

- If the $n \times 1$ vector $\mathbf{z}$ is multinormal with $N(\mathbf{0}, \mathbf{I})$, then we say that $\mathbf{z}$ is a standard normal vector.
- Then $z_1, \dots, z_n$ are pairwise independent standard normal random variables.
- The multinormal pdf simplifies to $f(\mathbf{z}) = (2\pi)^{-n/2} \exp(-\mathbf{z}'\mathbf{z}/2) = \prod_{i=1}^n \exp((-z_i^2/2)\sqrt{2\pi})$ which is the product of n independent standard normal pdf's.
- We restate here some results from the Review of probability and statistics in the next slide.

Chi-square distribution with n degrees of freedom $\chi^2(n)$

If $w = \mathbf{z}'\mathbf{z}$ where $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$ then $w \sim \chi^2(n)$. This is, the sum squares of n independent standard normal random variables has the chi-square distribution with parameter n .

F distribution with m and n degrees of freedom $F(m, n)$

If $v = \frac{(w_1/m)}{(w_2/n)}$, where $w_1 \sim \chi^2(m)$ and $w_2 \sim \chi^2(n)$ are independent, then $v \sim F(m, n)$. That is, the ratio of two independent chi-square variables, each divided by its degrees of freedom, has the F distribution, with degrees of freedom equal to the respective chi-squares.

Student's t -distribution

If $u = \frac{z}{\sqrt{w/n}}$ where $z \sim N(0, 1)$ and $w \sim \chi^2(n)$ are independent, then $u \sim t(n)$. That is, the ratio of a standard normal random variable to the square root of an independent chi-square random variable divided by its degrees of freedom has the Student's t -distribution

Remark

Observe that if $u \sim t(n)$, then $u^2 \sim F(1, n)$.

Quadratic Forms in Normal Vectors

- Now we are ready to state the important results from this section.
- All of the results in this section come without proof because they require more in-depth knowledge about matrix and linear algebra.¹

Q1

Suppose that the $n \times 1$ vector $\mathbf{y} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Let $w = (\mathbf{y} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \boldsymbol{\mu})$. Then $w \sim \chi^2(n)$.

Q2

Suppose that the $n \times 1$ vector $\mathbf{u} \sim N(\mathbf{0}, \mathbf{I})$. Let $\mathbf{M}$ be a nonrandom $n \times n$ idempotent matrix with $\text{rank}(\mathbf{M}) = r \leq n$, i.e., $\mathbf{M}$ doesn't have full rank. Let $w = \mathbf{u}' \mathbf{M} \mathbf{u}$. Then $w \sim \chi^2(r)$.

Q3

Suppose that the $n \times 1$ vector $\mathbf{u} \sim N(\mathbf{0}, \mathbf{I})$. Let $\mathbf{M}$ be a nonrandom $n \times n$ idempotent matrix with $\text{rank}(\mathbf{M}) = r \leq n$. Let $\mathbf{L}$ be a nonrandom matrix such that $\mathbf{L}\mathbf{M} = \mathbf{O}$, where $\mathbf{O}$ is a matrix of only zeros. Let $\mathbf{t}_1 = \mathbf{M}\mathbf{u}$ and $\mathbf{t}_2 = \mathbf{L}\mathbf{u}$. Then $\mathbf{t}_1$ and $\mathbf{t}_2$ are independent random vectors.

¹Namely, the procedure to diagonalize a matrix.